

9.1.1 Relating diagonalization to eigenvalues and eigenvectors

You may want to start your exploration of eigenvalues and eigenvectors by watching the video

- Eigenvectors and eigenvalues | Essence of linear algebra, chapter 14 from the 3Blue1Brown series. (We don't embed the video because we are not quite sure that the rules about doing so are.)

Here are the insights from that video in the terminology of this week.

9.1.1 Relating diagonalization to eigenvalues and eigenvectors

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Homework 9.1.1.1. Eigenvalues and eigenvectors are all about finding scalars, λ , and nonzero vectors, x , such that

$$Ax = \lambda x.$$

To help you visualizing how a 2×2 real-valued matrix transforms a vector on the unit circle in general, and eigenvectors of unit length in particular, we have created the function `Assignments/Week09/matlab/showeig.m` (inspired by such a function that used to be part of Matlab). You may now want to do a "git pull" to update your local copy of the `Assignments` directory.

Once you have uploaded this function to Matlab, in the command window, first create a 2×2 matrix in array `A` and then execute `showeig(A)`.

Here are some matrices to try:

```
A = [ 2      0
      0 -0.5 ]

A = [ 2      1
      0 -0.5 ]

A = [ 2      1
      1 -0.5 ]

theta = pi/4;
A = [ cos( theta)  -sin( theta )
      sin( theta )   cos( theta ) ]

A = [ 2      1
      0      2 ]

A = [ 2      -1
      -1    0.5 ]

A = [ 2      1.5
      1    -0.5 ]

A = [ 2      -1
      1    0.5 ]
```

Can you explain some of what you observe?

▼ [Solution](#)

```
A = [ 2      0
      0 -0.5 ]

A = [ 2      1
      0 -0.5 ]

A = [ 2      1
      1 -0.5 ]
```

If you try a few different symmetric matrices, you will notice that the eigenvectors are always mutually orthogonal.

```
theta = pi/4;
A = [ cos( theta)  -sin( theta )
      sin( theta )   cos( theta ) ]
```

In the end, no vectors are displayed. This is because for real-valued vectors, there are no vectors such that the rotated vector is in the same direction as the original vector. The eigenvalues and eigenvectors of a real-valued rotation are complex-valued. Unless θ is an integer multiple of π .

```
A = [ 2      1
      0      2 ]
```

We will see later that this is an example of a Jordan block. There is only one linearly independent eigenvector associated with the eigenvalue 2. Notice that the two eigenvectors that are displayed are not linearly independent (they point in opposite directions).

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 0.5 \end{bmatrix}$$

This matrix has linearly dependent columns (it has a nonzero vector in the null space and hence 0 is an eigenvalue).

$$A = \begin{bmatrix} 2 & 1.5 \\ 1 & -0.5 \end{bmatrix}$$

If you look carefully, you notice that the eigenvectors are not mutually orthogonal.

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 0.5 \end{bmatrix}$$

Another example of a matrix with complex-valued eigenvalues and eigenvectors.